SC Honors Geometry

Sub Day: Transformations 4.1 – 4.3

Name: _____ Period: ___ Partner: _____

Today you work on your own or with your partner. Every part has a way to check yourself: a gold ★ on a graph, a code letter, or a check sum. If a check fails, find the mistake before you move on. That's the whole point.

Rule card — use it!

Translation ⟨a, b⟩: (x, y) → (x + a, y + b)

Reflections: x-axis (x, −y) y-axis (−x, y) y = x (y, x) y = −x (−y, −x) x = h (2h − x, y) y = k (x, 2k − y)

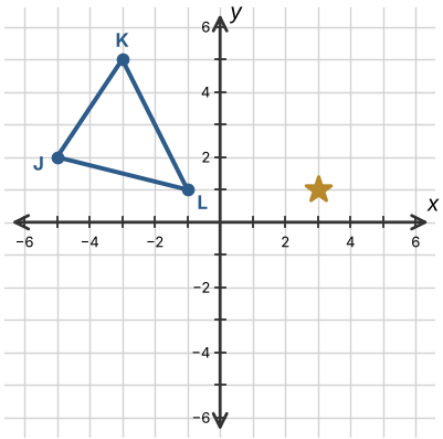
Rotations about the origin (counterclockwise): 90° (−y, x) 180° (−x, −y) 270° (y, −x)

Glide reflection: a translation, then a reflection in a line parallel to the translation. Do one step at a time.

Rigid motion: translations, reflections, and rotations keep every length and angle, so the image is congruent to the preimage.

Part A — Translations and reflections (4.1 – 4.2)

1. Graph △JKL with J(−5, 2), K(−3, 5), L(−1, 1) and its image after a translation by ⟨6, −4⟩. Label J′, K′, L′.



✓ **Check yourself:** If you are right, K′ lands exactly on the gold ★.

2. Write a rule for the translation that maps P(−2, 7) to P′(3, 1).

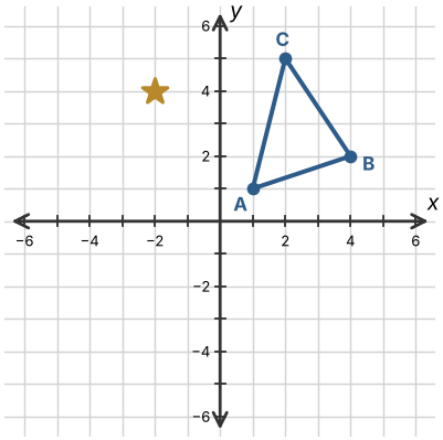
3. Reflect the point (−4, 6) in each line. (a) x-axis (b) y-axis (c) y = x (d) y = −x

4. Reflect M(5, −2) in (a) the line x = 2 (b) the line y = 1.

Find each answer in the code bank on page 3 and write its letter in the riddle.

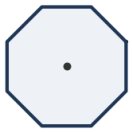
Part B — Rotations (4.3)

5. Graph △ABC with A(1, 1), B(4, 2), C(2, 5) and its image after a 90° rotation about the origin. Label A′, B′, C′.



✓ **Check yourself:** If you are right, B′ lands on the gold ★.

Regular octagon for #8:



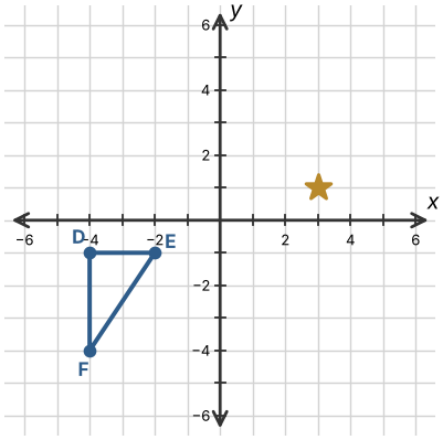
6. Rotate Q(3, −5) about the origin by (a) 90° (b) 180° (c) 270°.

7. A rotation about the origin maps every point (a, b) to (b, −a). How many degrees counterclockwise is it?

8. A regular octagon is rotated about its center. What is the smallest angle that maps it onto itself? How many lines of symmetry does it have?

Part C — Mixed practice (4.1 – 4.3)

9. Glide reflection: graph △DEF with D(−4, −1), E(−2, −1), F(−4, −4). Translate it by ⟨5, 0⟩, then reflect the image in the x-axis. Label the final image D″, E″, F″.



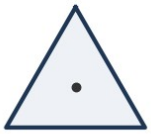
✓ **Check yourself:** If you are right, E″ lands on the gold ★.

10. A rotation about the origin maps every point (a, b) to (−a, −b). How many degrees is it?

11. Reflect R(2, 3) in (a) the line x = 1 (b) the line y = −2.

12. After a 180° rotation about the origin, the image of point T is T′(4, −7). What was T?

13. An equilateral triangle is rotated about its center. What is the smallest angle that maps it onto itself? How many lines of symmetry does it have?



✓ **Check yourself:** Add your angles in #7, #8, #10, and #13, plus the number of lines of symmetry in #8 and #13. The total should be 626.

Riddle time

Find each answer from #2 – #4 and #6 in the bank. Write its letter above the problem number. Some bank answers are traps: they are what you get from a common mistake.

How does a geometry teacher order coffee?

#2	#3a	#3b	#3c	#3d	#4a	#4b	#6a	#6b	#6c

Answer	Letter	Answer	Letter	Answer	Letter	Answer	Letter
(−6, −4)	D	(−5, 3)	U	(4, −6)	O	(−3, −5)	Y
(5, 3)	I	(5, 4)	W	(6, 4)	C	(6, −4)	H
(5, 0)	N	(x+5, y−6)	W	(−5, −3)	T	(−1, −2)	T
(3, 5)	P	(−3, 5)	S	(9, −2)	R	(4, 6)	T
(−4, −6)	I	(−4, 6)	E	(−6, 4)	A	(x−5, y+6)	L

If your letters don't spell a real answer, the problem under the wrong letter needs another look.

★ Challenge: Spiral review (Chapters 1 – 3)

Mix of everything so far. The answers are in the scrambled bank at the bottom. Cross each one off when you find it.

S1. Find the midpoint and the length of AB with A(−3, 4) and B(5, −2).

S2. M is the midpoint of AB. AM = 3x + 4 and MB = 5x − 6. Find AB.

S3. ∠1 and ∠2 are supplementary. m∠1 = (4x + 10)° and m∠2 = (2x + 20)°. Find both angles.

S4. Parallel lines are cut by a transversal. Alternate interior angles measure (5x − 12)° and (3x + 20)°. Find the angle measure.

S5. Write the equation of the line through (2, −1) that is perpendicular to y = −(2/3)x + 4.

S6. Parallel, perpendicular, or neither? 4x − 2y = 6 and y = 2x + 9

S7. Write the converse of “If two angles are vertical, then they are congruent.” Is the converse true?

S8. ★ Boss level: Reflect the line y = 2x + 1 in the x-axis. Write the equation of the image line.

Scrambled answer bank (includes 5 traps)

72° (1, 1); 14 38 converse is false (1, 1); 10 y = (3/2)x − 4 perpendicular y = −2x − 1 parallel 110° and 70° 19 y = −(3/2)x − 4 68°